15. The Computational Theory of Mind

Start with motivations for CTM- quotes from Haugeland.

Reorient towards: is CTM possible? Focus on problem issues: emotion, consciousness, intelligence.

CTM Defined

CTM v.1: "The central tenet of the theory is that a mind is a computer." (McLaughlin 2004)

Not a metaphor: the brain is literally a computer.
The mind is a computer just the way the heart is a pump. That is its actual biological function. But this is a philosophical conjecture. In cognitive science, computationalism is treated more like a methodological assumption than a world-view.

The brain is a natural computer.
The technical concept of a "computer" is that of a system whose function is to perform cognitive tasks (rational transitions) that relate cognitive contents by mechanically manipulating symbols which function to represent those contents— i.e. an information processor. This definition is neutral about the physical composition of the system or its history. The brain is a computer generated by natural selection.

Is it the brain or the mind?
Really, this is a claim about the brain, since computers are *embodied* systems. Instead, the mind is better thought of as what the brain *does*. Perhaps we should say, the mind *emerges* from a computer (the brain).

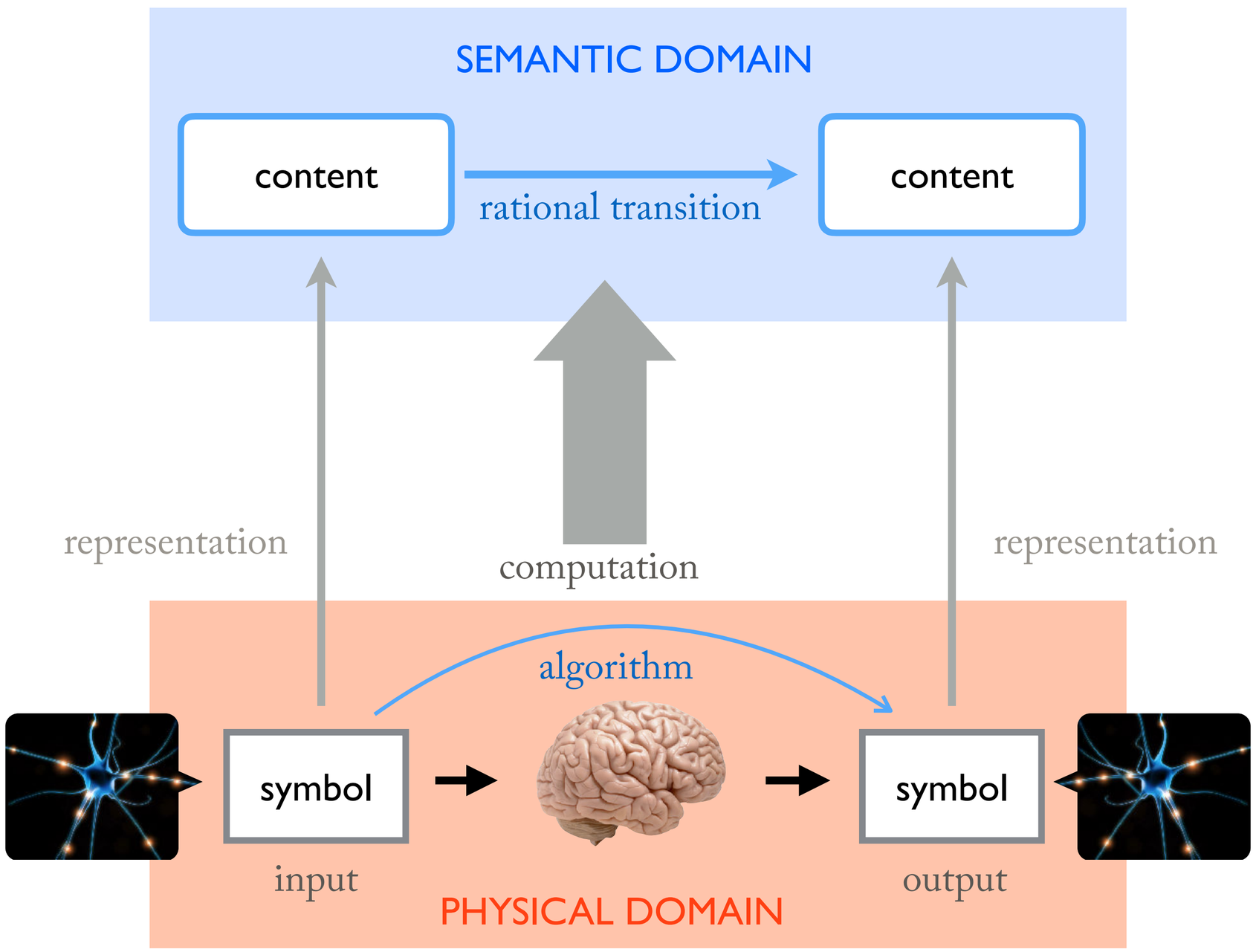
Is it a *computer* or a *computer program*?
If the brain is a *computer*, perhaps the mind is a *computer program*? This might be half right, in the sense that a computer executes (or performs) a computer program. But as we'll see, the concept of a "program" refers specifically to instructions that are distinct from the hardware they are stored in. But much of the mind emerges from biological hardware. The more general concept is that of an *algorithm*.

Is this a claim about all aspects of the mind, including consciousness and emotion?
While consciousness may be a computational state of the brain, there are few computationalists who believe that computation can *explain* phenomenology. In the end, computationalism is really about the cognitive mind. Emotions are debated, but many see emotions as important computational states which give us dispositions to act.

What is meant by "the mind" anyway?
Given that we have many different kinds of mental states, which are constantly changing over time?the concept of *mind* is not a particularly clear one. It's probably better to talk about mental *capacities*, mental *states*, and mental *processes*. (The mind, roughly, is the unity of these elements.) The idea is that mental states are computational states of the brain, and mental processes are computational processes of the brain.

How could the brain function to be a computer, since no one designed a brain?
The brain wasn't designed, but it does have an overall *biological* function, and its parts have many specific biological functions. States of the brain function to carry information about the world (to represent); and structures in the brain function to process these states (or, compute).

Computationalism v. 2: the brain is a natural computer, or information processor. The biological function of the brain is to perform a range of cognitive functions by carrying out mechanical transformations on physical symbols which compute those functions. Cognitive states are computational states of the brain. Cognitive processes are computational processes of the brain.



Computational Theory of Mind: the brain computes cognitive functions by manipulating symbolic representations in the brain according to biologically grounded algorithms.

Computation beyond Combinatorial Circuits

1. **Multi-functionality.** Natural cognitive systems compute many different functions. Our CC machines computes only one.

2. **Learning.** Natural cognitive systems change their behavior in response to experience. The behavior of our CC machines is fixed for eternity.

3. **Non-arithmetical functions.** Natural cognitive systems perform non-arithmetical functions, including decision making, action-planning, prediction, navigation. What can a CC do beyond +1B?

4. **Logical and probabilistic inference.** Among non-arithmetical functions some employ language-like reasoning, especially logic and inference. +1B is stuck in a world of integers; can we extend CC's to logic?

5. **Perception and action.** Natural cognitive systems exist within an environment. They gather information from that environment (perception), and they change that environment (action). +1B has no obvious causal connection to the natural order. Can CC's perform perception?

6. **Biological basis.** Brains are built from neurons, not logic gates; neurons are put together to form neural nets, not combinatorial circuits. So CC's fails to capture the physical and functional basis of natural computation.

7. **Emotion and consciousness.** CCs are incapable of emotion or consciousness. (Right?) It is not clear, from the current standpoint, what would even be required to achieve this.

8. **Original functionality.** Natural cognitive systems have the biological function of performing certain information-processing tasks, as a result of natural selection. The +1B machine's only function was stipulated in a problem set.

Further Notes

- How CTM answers Leibniz Challenge
  Analogy with computers ≈ the brain is a computer, the mind is an algorithm.
- Computers are purely physical.
- They have thought-like states, and intelligence-like processes.
- They can cause and be caused by physical events.
- The physical domain and the semantic domain
- Natural and artificial computers.

- **CTM and Cognitive Science**
  - Cognitive science isn't philosophically committed.
  - But it relies on CTM as a working hypothesis.
  - CTM is a conjecture, not a conclusion.